

Privacy and Security in the Digital Age: Balancing Information Protection and Individual Privacy

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Executive Summary

Organizations increasingly depend on digital systems to collect, store, and share personal information. This dependence creates both privacy and security responsibilities. Privacy concerns how information is collected and used, while security focuses on protecting information and systems from unauthorized access or misuse. Although these responsibilities are closely related, security measures can sometimes create additional privacy concerns.

This report examines research on privacy and security, including human behavior, security controls, organizational responsibility, and stakeholder trust. The research indicates that organizations should use technical safeguards such as encryption, multifactor authentication (MFA), access controls, and monitoring while also applying privacy practices such as data minimization, transparency, and appropriate information retention. Effective protection requires more than technology; it requires people, policies, and responsible organizational decision-making.

Overall, privacy and security should be treated as complementary responsibilities. Privacy should only be limited when there is a legitimate and necessary security purpose, and the response should be proportionate to the identified risk. Organizations that integrate privacy into their security programs can protect sensitive information while maintaining individual privacy and stakeholder trust.

Introduction

Digital technology has transformed how organizations collect and manage information. Customers, employees, and organizations routinely provide and store personal information through websites, mobile applications, databases, and other digital systems. This information can

include financial records, employment information, location data, and identifying information. As the amount of stored information increases, so does the potential impact of cyberattacks, phishing, ransomware, stolen credentials, and data breaches.

Privacy and security address different aspects of this problem. Privacy concerns whether personal information is collected, used, shared, and retained appropriately. Bélanger and Crossler (2011) explain that information privacy involves individuals, organizations, technology, and institutional practices. Security focuses on protecting information and systems from unauthorized access, disclosure, modification, or destruction. NIST (2020b) provides security and privacy controls that organizations can use to reduce these risks.

The relationship between privacy and security becomes especially important when organizations collect personal information to improve security. For example, monitoring systems, cameras, access logs, and location tracking may help identify threats, but can also collect information about individuals that is not necessary for the original purpose. Nissenbaum (2004) argues that privacy depends partly on whether information flows appropriately within a particular context. Therefore, organizations must consider both whether information is secure and whether its collection and use are appropriate.

This report examines how organizations can balance these responsibilities. The central argument is that privacy should be limited only when there is a legitimate, necessary, reasonable, and proportionate security need. Organizations should collect only the information needed, protect it appropriately, and use the least intrusive and most effective security measures.

Methodology

This report uses a qualitative literature review of ten scholarly and authoritative sources addressing information privacy, cybersecurity, human behavior, organizational practices, and trust. The sources include peer-reviewed journal articles, legal scholarship, information systems research, and publications from NIST.

The research was examined for common themes involving privacy expectations, security risks, human behavior, technical controls, organizational responsibility, and trust. Acquisti et al. (2015), Bélanger and Crossler (2011), and Pavlou (2011) provide research on privacy behavior and information systems. Nissenbaum (2004) and Solove (2006) provide frameworks for understanding privacy. Martin (2018) and Xu et al. (2011) address privacy and trust, while Cavoukian (2010) and NIST (2020a, 2020b) provide practical approaches for incorporating privacy and security into organizational systems.

Findings

The research demonstrates that privacy and security are closely connected but should not be considered the same responsibility. Security protects information from unauthorized activity, while privacy addresses whether information should be collected and how it should be used.

Solove (2006) identifies several types of privacy problems, including information collection, processing, dissemination, and invasion. This demonstrates that an organization can have strong cybersecurity while still violating privacy through excessive collection or inappropriate use of information. For example, an organization may securely store employee information but collect more information than necessary for its stated purpose.

Nissenbaum (2004) further explains that privacy depends on the context in which information is shared. Information provided for one purpose does not automatically give an

organization unlimited permission to use it for another purpose. This distinction is important when organizations implement security technologies that collect personal information.

Human behavior adds another layer to the problem. Acquisti et al. (2015) found that individuals' privacy decisions are influenced by factors such as convenience, limited knowledge, and difficulty predicting how information may be used. Similar human factors affect cybersecurity. Employees may fall for phishing attacks, use weak passwords, or unintentionally disclose sensitive information.

Organizations should therefore address both privacy and human security risks. Employee training can help workers recognize phishing attempts, protect credentials, handle sensitive information, and report suspicious activity. However, organizations should not rely on training alone. Technical controls should reduce the potential impact of human mistakes.

Organizations need multiple layers of security because no single safeguard can eliminate every threat. NIST (2020b) identifies controls addressing access management, authentication, incident response, risk assessment, system integrity, and contingency planning. Encryption, MFA, least-privilege access, monitoring, and reliable backups can support these controls.

Encryption can protect information from unauthorized access, while MFA can reduce the risk associated with stolen passwords. Least-privilege access limits users to information necessary for their responsibilities. Monitoring can help detect suspicious activity, and backups can reduce the effects of ransomware or system failure.

However, security controls can also create privacy risks. Monitoring systems, for example, may collect extensive information about employee or customer behavior. If organizations collect or retain information beyond what is necessary, a security measure can

become a source of privacy risk. Security controls should therefore be proportional to the threat they are intended to address.

Privacy should be considered when systems are designed rather than after problems occur. Cavoukian's (2010) Privacy by Design principles emphasize proactive privacy protection and the integration of privacy into system design. Data minimization is an important part of this approach. Organizations should collect information only when necessary for a legitimate purpose, not simply because technology allows it.

The NIST Privacy Framework also encourages organizations to manage privacy as part of enterprise risk management (NIST, 2020a). Organizations should consider why information is collected, who needs access, how long it will be retained, and what risks collection creates. This approach can reduce privacy exposure while still allowing organizations to address legitimate security threats.

Privacy practices can have significant effects on organizational trust. Martin (2018) found that privacy violations can negatively affect trust online. When individuals believe that an organization has mishandled their information, they may become less willing to provide information or continue using the organization's services.

Xu et al. (2011) similarly found that institutional privacy assurances can influence individual privacy concerns. Clear policies, meaningful safeguards, and responsible information practices can increase confidence in an organization. In contrast, unclear or unexpected information practices can increase concern.

Pavlou (2011) emphasizes that privacy research involves relationships among individuals, organizations, and technology. This means that privacy is not simply a technical

issue. It is also an organizational and human issue. Organizations must consider how their information practices affect customers, employees, and other stakeholders.

Bélanger and Crossler (2011) demonstrate that information privacy depends on both the individual expectations and organizational practices, supporting the argument that privacy cannot be addressed through technical controls alone. These findings support the need for a balanced approach rather than treating privacy and security as competing objectives.

Organizations should first identify the security risk they are attempting to address. They should then determine whether collecting or monitoring personal information is necessary to address that risk. If necessary, the organization should use the least intrusive and most effective approach. Access should be limited, unnecessary information should not be collected, and information should not be retained longer than necessary.

When these principles are applied together, privacy and security can reinforce one another. Data minimization reduces the amount of information available to attackers; access controls reduce unnecessary exposure; encryption protects sensitive information; and MFA strengthens authentication. Employee training and organizational policies address human behavior and establish expectations for responsible use of information.

An effective privacy and security program, therefore, requires people, policies, and technology. Organizations need employees who understand their responsibilities, leaders who support appropriate practices, policies that establish clear expectations, and technical controls that protect information. Continuous risk assessment is also necessary because both technology and cyber threats continue to change.

Conclusion

Privacy and security are essential responsibilities for organizations operating in a digital environment. Although they address different concerns, both are necessary to protect sensitive information. Security protects information from unauthorized access and other threats, while privacy ensures that information is collected and used appropriately.

The research demonstrates that organizations cannot rely on technology alone. Encryption, MFA, access controls, monitoring, and backups are important security measures, but employee training, policies, risk assessments, and responsible data-management practices must support them. Privacy by Design and the NIST Privacy Framework demonstrate the importance of considering privacy during system and process development rather than waiting until after a violation occurs (Cavoukian, 2010; NIST, 2020a).

Organizations must also recognize that privacy affects stakeholder trust. Privacy violations can damage relationships with customers and employees and create reputational consequences (Martin, 2018). For this reason, privacy should only be limited when there is a legitimate and necessary security purpose. Even when a security need exists, organizations should collect the minimum information required and use measures that are reasonable and proportionate to the risk.

Ultimately, privacy and security do not have to compete. A balanced approach that combines people, policies, and technology can protect organizational information while respecting individual privacy. As organizations continue to rely on digital technology, maintaining this balance will remain essential for reducing security risks, protecting sensitive information, and preserving stakeholder trust.

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